

Financials Cloud – Technical | Workshop Agenda

Day 1 Approvals

- ✓ Configure using Spreadsheet
- ✓ Sample Use Cases
- ✓ Migration, Archive & Purge
- ✓ Advanced Mode
- ✓ Best Practices
- ✓ Configurable Notifications
- ✓ Transactions Console

Day 2 & 3 Reports

- ✓ Financial Reporting Studio
- ✓ SmartView
- ✓ OTBI
- ✓ Sunburst
- ✓ BI Publisher

Day 3 Integrations

- ✓ File Based Data Import
- ✓ Financial Extracts
- ✓ BIP Report Extract
- ✓ BI Cloud Extracts
- ✓ REST & SOAP API
- ✓ B2B Messaging
- ✓ Data Conversion Considerations
- ✓ File Encryption & Decryption
- ✓ API Authentication
- ✓ Business Events

Day 4 Extensibility

- ✓ Sandbox
- ✓ Appearance
- ✓ Structure
- ✓ Page Composer
- ✓ Page Integration
- ✓ User Interface Text
- ✓ Messages, Lookup & Data Security
- ✓ Personalizations
- ✓ Flexfields



ORACLE

Oracle Financials Cloud

Integrations

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Oracle Development



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Agenda

- **File Based Data Import**
- Financial Extracts
- BIP Report Extract
- BI Cloud Extracts
- REST & SOAP API
- B2B Messaging
- Data Conversion Considerations

Process Flow | Discussion & Demo

Use Case Scenarios

- High volume recurring data imports
- One-time master data loading
- Open transactions import as part of cut-over
- Syncing of common objects in co-existence scenarios

Process Flow

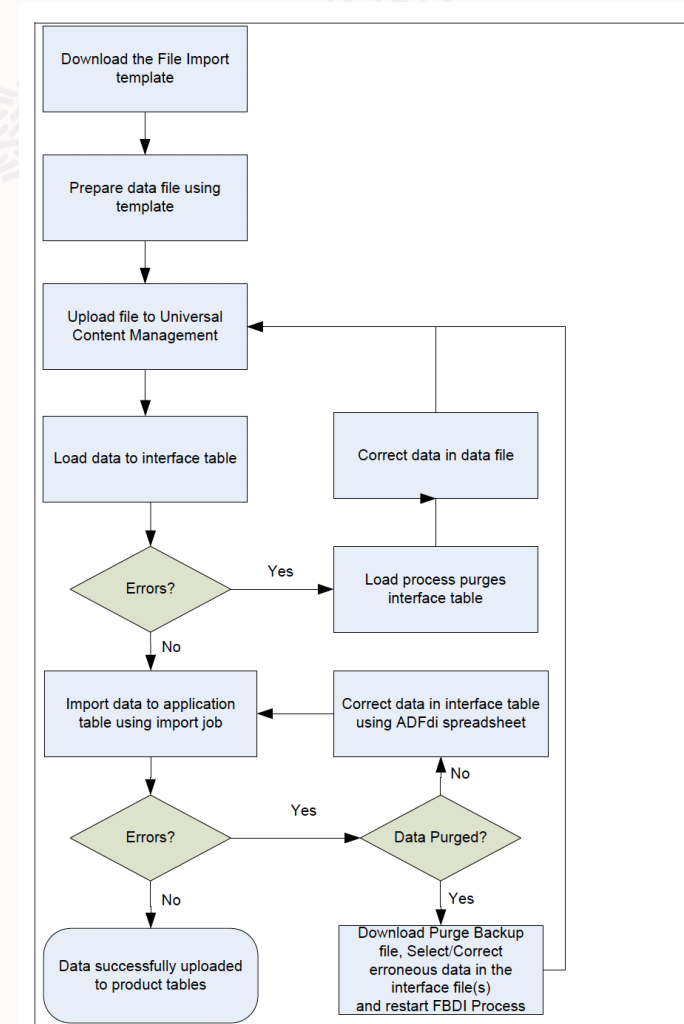
- Download object specific [FBDI template](#)
- Fill up the template and prepare the data file
- Upload the datafile to UCM
- Submit 'Load Interface File for Import' process
- Submit object specific import process; for example: 'Import Journals' process
- Download the output & log files for review

Security

- **Integration Specialist** job role; inherits project specific integration specialist roles, UCM access & REST API service calls
- **AttachmentsUser** abstract role(for UCM access); inherited by Employee and Contingent Worker roles

Demo Flow

- Bulk load GL Journals



- [How data is processed in each Financials FBDI process](#)
- `FND_FILE_UPLOAD_MAX_SIZE` UCM file size (2GB)

Error Correction & Purge

Errors due to structural issues or formatting of file:

- Review the uploaded error logs
- Change any structural or formatting anomalies in the data
- Generate the zip file containing the CSV files using the respective import template
- Upload the corrected file to the UCM server and resubmit the Load Interface File for Import process
- Repeat these steps until the process successfully loads all the data

Errors due to invalid data:

- Review the errors in the import log
- If auto purge is NOT enabled in your import process, then:
 - Correct the error records using the applicable ADFdi correction spreadsheets.
- If auto purge is enabled in your import process, then use these steps:
 - Download the purge erroneous zip file from the File Import and Export page.
 - Select the erroneous data records from the interface file and correct them.
 - Follow the FBDI process to resubmit the corrected data.

Purge Process:



Name Purge Interface Tables

Description Purge data from interface tables pertaining to ... ☐ Notify me when this process ends

Schedule As soon as possible **Submission Notes**

Basic Options

Parameters

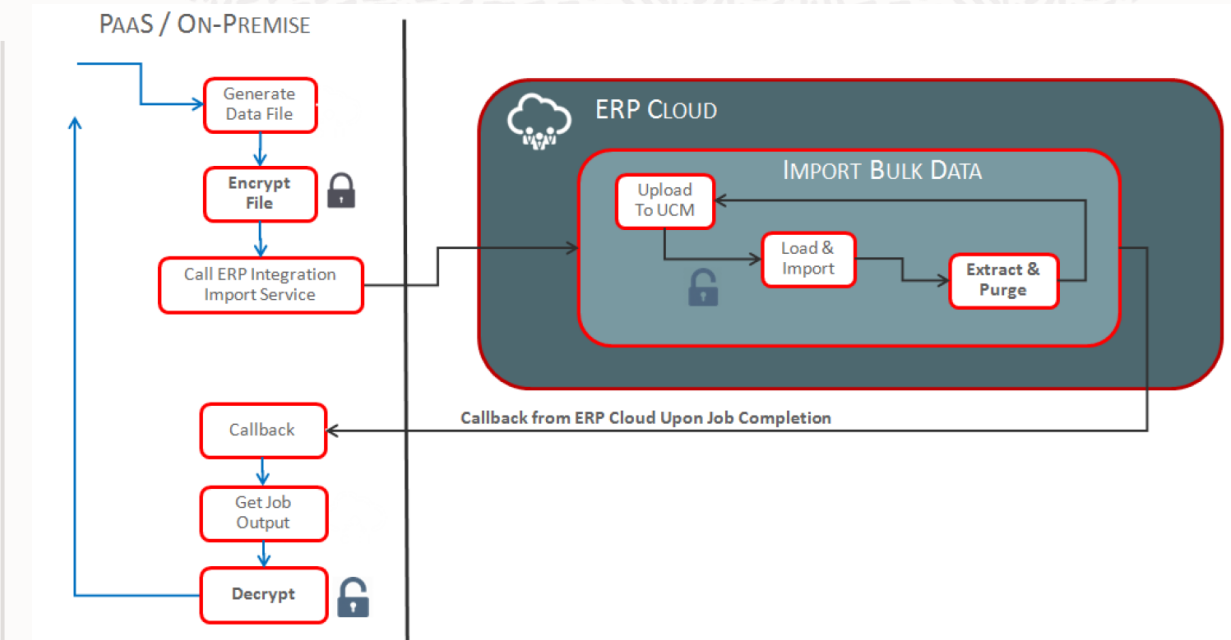
Purge Process Intent File-based data import ▼

Import Process

Automation using REST API | Overview

Automation Flow

- Generate the data file (zipped csv file) programmatically at the source system
- Optionally encrypt the file using PGP encryption [Appendix J]
- Format of the data file should conform to FBDI format
- Use **ERP Integration Service** REST API to programmatically execute the subsequent steps:
 - Upload the file to UCM
 - Submit 'Load Interface File for Import' process
 - Monitor the progress of the process (or) register the callback service
 - Submit 'Import Journals' process
 - Monitor the progress of the process (or) register the callback service
 - Download the output & log files
- Decrypt the files [Appendix J]



- [Using External Data Integration Services for Oracle ERP Cloud \(2102800.1\)](#)
- [Blog - Oracle ERP Cloud Integration Services - Import Data](#)

Automation using REST API | Multi-Step Approach | Demo

[REST API URI](#) `https://<server-name>/fscmRestApi/resources/11.13.18.05/erpintegrations`

(1) **UploadFileToUCM** [uploads the data file base-64 content into UCM folder; returns **DocumentId**]

Sample request payload:

```
{
  "OperationName": "uploadFileToUCM",
  "DocumentContent": "UESDBBQAAAAIACBaK1G36cyudgAAAKIBAAAPAAAAAR2xJbnRl",
  "DocumentAccount": "fin$/generalLedger$/import$",
  "ContentType": "zip",
  "FileName": "G1InterfaceDemo_0206.zip",
  "DocumentId": null
}
```

DocumentContent
Refer [Base64](#)

DocumentAccount
Refer Appendix A

(2) **SubmitESSJobRequest** [submits 'Load Interface File for Import' process for the document uploaded in prior step; returns **RequestId**]

Sample request payload:

```
{
  "OperationName": "submitESSJobRequest",
  "JobPackageName": "oracle/apps/ess/financials/commonModules/shared/common/interfaceLoader",
  "JobDefName": "InterfaceLoaderController",
  "ESSParameters": "15,2737430,N,N"
}
```

15 : ERP_INTERFACE_OPTIONS_ID
JobDefName & JobPackageName
Refer Appendix A

ESSParameters
Refer Appendix B

(3) **GetEssJobStatus** [get the status of the job submitted in the prior step; returns the **status**]

Sample URI:

`https://<server-name>/fscmRestApi/resources/11.13.18.05/erpintegrations?finder=ESSJobStatusRF;requestId=2745786`



Automation using REST API | Multi-Step Approach | Demo

[REST API URI](#) https://<server-name>/fscmRestApi/resources/11.13.18.05/erpintegrations

(4) **SubmitESSJobRequest** [submits 'Import Journals' process for the for specific ledger & other parameters; returns **RequestId**]

Sample request payload:

```
{
  "OperationName": "submitESSJobRequest",
  "JobPackageName": "/oracle/apps/ess/financials/generalLedger/programs/common",
  "JobDefName": "JournalImportLauncher",
  "ESSParameters": "300000206228233,Other,300000206229310,1234567,N,N,N"
}
```

JobDefName & JobPackageName
Refer Appendix A

ESSParameters
Refer Appendix B

(5) **GetEssJobStatus** [get the status of the job submitted in the prior step; returns the **status**]

Sample URI:

https://<server-name>/fscmRestApi/resources/11.13.18.05/erpintegrations?finder=ESSJobStatusRF;requestId=2745793

(6) **DownloadESSJobExecutionDetails** [downloads the output & logfiles of the submitted job]

Sample URI:

https://<server>/fscmRestApi/resources/11.13.18.05/erpintegrations?finder=ESSJobExecutionDetailsRF;requestId=2745793



Automation using REST API | Single-Step Approach | Demo

[REST API URI](#) `https://<server-name>/fscmRestApi/resources/11.13.18.05/erpintegrations`

(1) **ImportBulkData** [uploads file, submits both the imports; returns **RequestId**]

Sample request payload:

```
{
  "OperationName": "importBulkData",
  "DocumentContent": "UESDBBQAAAAIACBaK1G36cyudgAAAKIBAAAPAAAA",
  "ContentType": "zip",
  "FileName": "GInterfaceDemo_020602.zip",
  "DocumentAccount": "fin$/generalLedger$/import$",
  "JobName": "/oracle/apps/ess/financials/generalLedger/programs/common,JournalImportLauncher",
  "ParameterList": "300000206228233,Other,300000206229310,1234567,N,N,N",
  "CallbackURL": "#NULL",
  "NotificationCode": "10",
  "JobOptions": "InterfaceDetails=15,ImportOption=Y,PurgeOption=Y,ExtractFileType=ALL"
}
```

DocumentContent

Refer [Base64](#)

JobName & DocumentAccount

Refer Appendix A

ParameterList

Refer Appendix B

NotificationCode

Refer Appendix C

CallbackURL

Refer Appendix D

ExtractFileType

ALL / OUT / LOG

(2) **GetEssJobStatus** [get the status of the job submitted in the prior step; returns the **status**]

Sample URI:

`https://<server-name>/fscmRestApi/resources/11.13.18.05/erpintegrations?finder=ESSJobStatusRF;requestId=2745793`

(3) **DownloadESSJobExecutionDetails** [downloads the output & logfiles of the submitted job]

Sample URI:

`https://<server>/fscmRestApi/resources/11.13.18.05/erpintegrations?finder=ESSJobExecutionDetailsRF;requestId=2745793`



Considerations

To minimize the risks of an unsuccessful data load, ensure the following:

- Unused columns can be hidden, but they cannot be reordered or deleted
- Deleting or reordering columns causes the load process to fail and results in an unsuccessful data load
- External data must conform to the data types accepted by the control file and process for the associated database column.
- Date column values must appear in the YYYY/MM/DD format.
- Amount column values can't have separators other than a period (.) as the decimal separator.
- Negative values must be preceded by the minus (-) sign.
- Column values that require whole numbers include data validation to allow whole numbers only
- For columns that require internal ID values, refer to the bubble help text for additional guidance about finding these values





Agenda

- File Based Data Import
- **Financial Extracts**
- BIP Report Extract
- BI Cloud Extracts
- REST & SOAP API
- B2B Messaging
- Data Conversion Considerations

Process Flow | Discussion & Demo

Use Case Scenarios

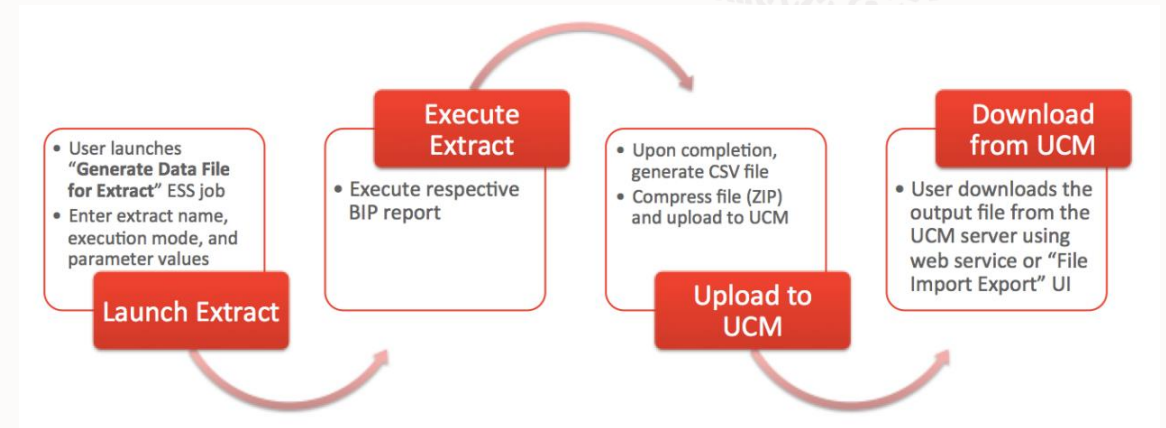
- High volume incremental recurring data extracts
- One-time master or transaction data extract
- Extract the data to integrate with downstream applications
- Export the data for external analysis

Process Flow

- User launches “Generate Data File for Extract” job with respective parameter values and execution mode.
- ERP Cloud executes the respective BIP report.
- Upon completion, a CSV file is generated in a compressed format and uploaded to UCM.
- User downloads the file using web services
- Role – Financials Integration Specialist

Demo Flow

- Payables Transactions Extract



- [Using Seeded Data Extract Services for Oracle ERP Financial Cloud](#) (Doc ID 2303683.1)
- [ERP - Introduction to Financial Seeded Data Extracts](#)

Automation using REST API | Discussion & Demo

[REST API URI](https://<server-name>/fscmRestApi/resources/11.13.18.05/erpintegrations) https://<server-name>/fscmRestApi/resources/11.13.18.05/erpintegrations

ParameterList
Refer Appendix B

NotificationCode
Refer Appendix C

CallbackURL
Refer Appendix D

ExtractFileType
ALL / OUT / LOG

(1) **ExportBulkData** [submits payables extract process; returns **RequestId**]

Sample request payload:

```
{
  "OperationName" : "exportBulkData",
  "JobName": "/oracle/apps/ess/financials/commonModules/shared/common/outbound,FinOutboundProcess",
  "ParameterList": "92,/oracle/apps/ess/financials/commonModules/shared/common/outbound;PayablesTransactionsExtract,BIP
REPORT,FULL_EXTRACT,FullExtractPrefix_,300000206233222,#NULL,#NULL,#NULL,#NULL,#NULL,Jun-
21,N,N,300000206229310,#NULL,#NULL,#NULL,FULL_EXTRACT,#NULL,#NULL,#NULL,PayablesTransactionsExtract,#NULL",
  "CallbackURL" : "#NULL",
  "NotificationCode": "10",
  "JobOptions": "ExtractFileType=ALL"
}
```

(2) **GetEssJobStatus** [get the status of the job submitted in the prior step; returns the **status**]

Sample URI:

https://<server-name>/fscmRestApi/resources/11.13.18.05/erpintegrations?finder=ESSJobStatusRF;requestId=2745793

(3) **DownloadESSJobExecutionDetails** [downloads the output & logfiles of the submitted job]

Sample URI:

https://<server>/fscmRestApi/resources/11.13.18.05/erpintegrations?finder=ESSJobExecutionDetailsRF;requestId=2745793



Customize Seeded Extracts | Discussion & Demo

[Using Seeded Data Extract Services for Oracle ERP Financial Cloud](#) (Doc ID 2303683.1)

- Run 'Generate Data File for Export' process
- Choose Extract Mode = 'Register custom enterprise scheduler job'
- Provide your own 'User-Defined Job Name' and Submit
- This will create a custom ESS Job Definition, which can be queried using 'Define Enterprise Scheduler Job Definitions and Job Sets for Financials' task
- ESS job is of 'BIPJobType' and points to '/Custom/Financials/<user-defined-job-name>.xdo'
- Create a copy of the original .xdo and .xdm files in Custom folder under BI Catalog. Names should exactly match with your <user-defined-job-name>.
- Following payload can be used to invoke your custom job:

```
{
  "OperationName" : "exportBulkData",
  "JobName": "/oracle/apps/ess/financials/commonModules/shared/common/outbound,FinOutboundProcess",
  "ParameterList": "#NULL,/Custom/Financials/PTTPayablesExtract.xdo,BIPONLY,#NULL,PTTPayablesExtract,comma-separated report params",
  "CallbackURL" : "#NULL",
  "NotificationCode": "10",
  "JobOptions": "ExtractFileType=ALL"
}
```





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Custom BIP Report Extract | Discussion & Demo

Process Flow

- Create a custom BIP Report
- Create a custom Enterprise Scheduler Job to invoke the custom BIP Report (Define Enterprise Scheduler Job Definitions and Job Sets for Financials task)
- Submit it from Tools > Scheduled Processes workarea
- Alternatively schedule the job to run at fixed interval

Demo Flow

- Define and execute the custom job 'Custom GL Code Combinations Extract'

- ✓ [Using External Data Integration Services for Oracle ERP Cloud](#)
- ✓ [Tables and Views for Financials](#)
- ✓ Appendix F: Custom ESS Job Parameters

The screenshot shows the 'Job Definition' form in Oracle Enterprise Scheduler. The form is divided into two main sections: 'Job Definition' and 'Job Properties'.

Job Definition Section:

- Display Name:** Custom GL Code Combinations ESS Job
- Name:** CUSTOMGLCC
- Path:** /PTS/
- Application:** (Dropdown menu)
- Description:** (Text area)
- Retries:** (Text input)
- Job Category:** (Text input)
- Timeout Period:** (Text input)

Job Properties Section:

- Job Application Name:** FscmEss
- Enable submission from Enterprise Manager:** (Unchecked checkbox)
- Job Type:** BIPJobType
- Class Name:** oracle.xdo.service.client.scheduler.BIPJobExecutable
- Default Output Format:** XML
- Report ID:** /Custom/FIN/GL_CC_REF
- Priority:** (Text input)
- Allow Multiple Pending Submissions:** (Unchecked checkbox)
- Enable submission from Scheduled Processes:** (Checked checkbox)

Parameters Section:

The 'CUSTOMGLCC: Parameters' section is expanded, showing a table of parameters:

Parameter Prompt	Data Type
Ledger Name	String

Secure the custom ESS job:

- Running 'Import User and Role Application Security Data' will create a custom privilege for the job definition.
- How to Grant Execution Privileges to ESS Custom Jobs? (Doc ID 2251561.1)



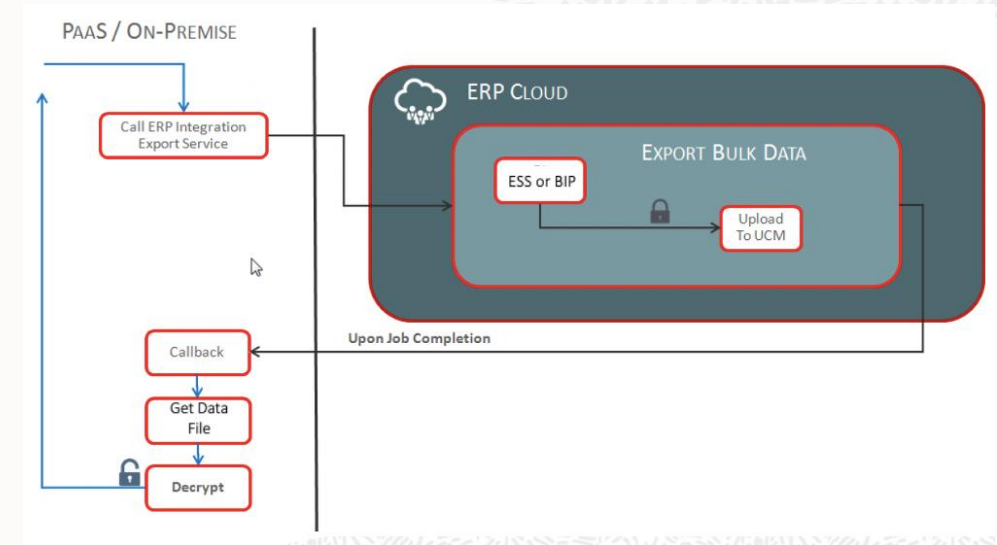
Custom BIP Report Extract | Automation using REST API | Demo

SubmitESSJobRequest

Sample request payload:

```
{
  "OperationName": "submitESSJobRequest",
  "JobPackageName": "/oracle/apps/ess/custom/PTS",
  "JobDefName": "CUSTOMGLCC",
  "ESSParameters": "InFusion Ledger US"
}
```

PTS : 'path' specified in the Job Definition
Refer to Job Definition in previous slide



- [Using External Data Integration Services for Oracle ERP Cloud](#) (2102800.1)
- [How to Configure PGP in BIP for FTP](#) (DocID 1987283.1) [Appendix J]



Agenda

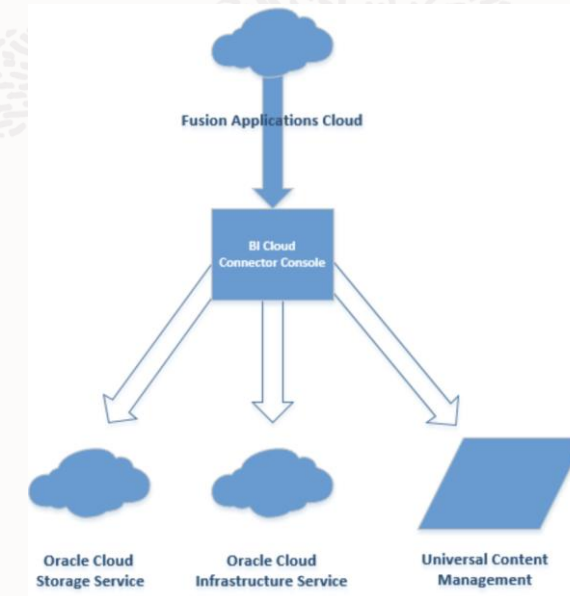
- File Based Data Import
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- **BI Cloud Extracts**
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BI Cloud Extracts

Use **BI Cloud Connector** to extract business intelligence and other data in bulk and load it into designated storage areas. Let's say you want to extract data for load into a warehouse for analysis. You determine which offerings and columns you want to extract, set the storage type and location for your extract, and run a one-time extract or incremental scheduled extracts.

You can easily:

- Extract data for your offerings in an easily-used format.
- Define your own offerings and associate data stores, or view objects, with them.
- Register custom objects.
- Refine the scope of columns in offering data stores included in a Cloud Extract.
- Export configured offerings and associated data stores.
- Use full or incremental extracts.
- Manage refresh metadata and specify dates for incremental refresh comparison.
- Specify where Cloud Extracts are stored.
- Schedule daily, weekly, monthly, or yearly extracts.
- Run an ad-hoc extract.
- Manage extracts as specific jobs, isolating incremental refresh metadata.
- Monitor extracts and review logs.



- BI Cloud Connector Console URL: <https://<server-name>/biacm>
- [Introduction](#) (CCC Replay)
- [Creating BI Cloud Extracts](#)
- [Extract Data Stores for Financials - Oracle Financials Cloud Extract Data Stores for Financials](#)
- [Database Lineage Mapping Spreadsheet \(VO – Table mapping\)](#)
- [Blog – download extraction files from SaaS](#)
- [Blog – view extracted files in UCM](#)
- [Best Practices](#) (Doc ID 2679006.1)
- [Automate using REST API](#)



Agenda

- File Based Data Import
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- **REST & SOAP API**
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REST API & SOAP API | Discussion & Demo

REST API

- [Financials REST API](#)
- [Use Cases & Sample Payloads](#)
- Sample Demo & Discussion
 - Payables Invoice GET & PATCH
 - Attach file to Invoice (POST)
- Supported: Basic Auth & Token Based (JWT & SAML)
- Refer to Appendix G
- [Self Service API Authentication Setup](#)
 - Inbound API Calls – JWT & SAML
 - Outbound API Calls – JWT Custom Claims & OAuth
- [Visual Builder Add-In for Excel](#)

SOAP API

- [Financials SOAP API](#)
- It is recommended that you utilize REST web services. If a REST web service is not available then utilize the SOAP web service associated with the component.
- [Oracle ERP Cloud Object Attachment Service](#)



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B2B Messaging | Collaboration Messaging Framework | Supported Messages

Procure to Pay

- Acknowledge Change Purchase Order Inbound
- Acknowledge Purchase Order Inbound
- Process Purchase Order Outbound
- Process Receipt Acknowledgment Inbound
- Process Invoice Inbound
- Process Shipment Inbound
- Process Credit Note or Memo
- Change Purchase Order Outbound
- Cancel Purchase Order Outbound
- Process Receipt Advice Outbound
- Acknowledge Rejected Invoice Outbound

Order to Cash

- Process Purchase Order Inbound
- Change Purchase Order Inbound
- Cancel Purchase Order Inbound
- Acknowledge Purchase Order Outbound
- Acknowledge Change Purchase Order Outbound
- Process Shipment Outbound
- Process Invoice Outbound

Supply Chain Operations

- Process Forecast Outbound
- Acknowledge Forecast Inbound

✓ To enable / disable, use Manage Collaboration Messaging Configuration task

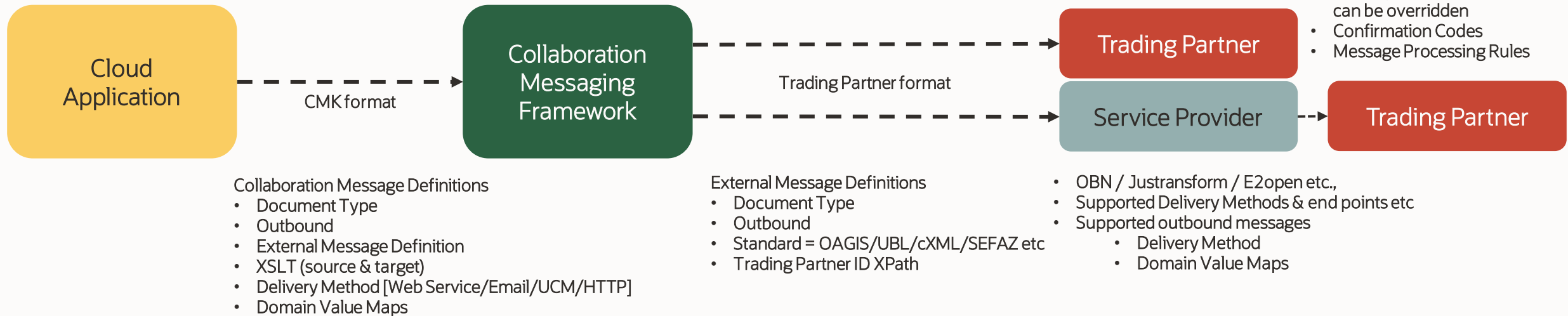
- [Electronic Invoicing in Oracle Payables Cloud](#)
- [B2B Messaging](#)
- [Oracle Procurement Cloud Purchasing Electronic Communication](#)
- Refer Appendix F for EDI Support



Collaboration Messaging Framework | Outbound Messaging

Associate Trading Partners with Suppliers and Customers

- Procure to pay > Manage Supplier B2B Configuration
- Supply chain operations > Manage Supplier B2B Configuration
- Order to cash, ordering and shipping at the customer level > Manage Customer Collaboration Configuration
- Order to cash, invoicing at the customer account level > Manage Customer Account Collaboration Configuration



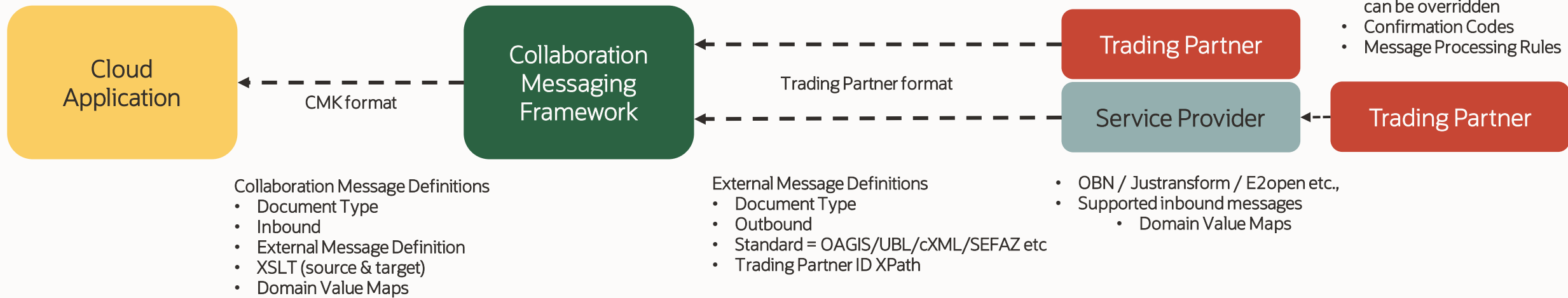
Collaboration Messaging Framework | Inbound Messaging

Associate Trading Partners with Suppliers and Customers

- Procure to pay > Manage Supplier B2B Configuration
- Supply chain operations > Manage Supplier B2B Configuration
- Order to cash, ordering and shipping at the customer level > Manage Customer Collaboration Configuration
- Order to cash, invoicing at the customer account level > Manage Customer Account Collaboration Configuration

- Trading Partner ID, ID Type
- Supported inbound messages
 - Domain Value Maps
- Confirmation Codes
- Message Processing Rules

- Trading Partner ID, ID Type
- Messages are inherited & can be overridden
- Confirmation Codes
- Message Processing Rules



- ✓ Collaboration Messaging Framework can receive B2B messages from trading partners or service providers directly using SOAP web services that are available to consumers.
- <https://host/soa-infra/services/default/CmkCollaborationMessagingOAGIS10InboundComposite/CollaborationMessageService> [payload be a CollaborationMessage]
- <https://host/soa-infra/services/default/CmkCollaborationMessagingInboundServiceComposite/CollaborationMessageServiceV2> [B2BMessage, accept any XML document]
- Basic Auth [CMK_INVOKE_INBOUND_COLLAB_DOC_SERVICE_PRIV role]





Agenda

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Data Conversion Considerations | Guidelines

Master Data

- Suppliers, Customers, Items, Bank Accounts etc

AP / AR Invoices

- Open invoices for unpaid balances only
- Try to summarize the lines and have only one line
- Un-applied prepayments; pay them using dummy bank account
- Open customer advances in AR; as unapplied or on-account receipt

Expense Reports

- Expense Reports under processing
- Open cash advances

Asset Balances

- Assets that are not retired in the source system
- Load into corporate book with accumulated depreciation details
- [Assets Data Conversion and Migration](#)
- Fusion Assets Data Conversion Suggestions (Doc ID 2052502.1)
- Fixed Asset Beginning Balances - Data Conversion (Doc ID 2011242.1)

GL Balances

- Trial balances loaded as GL journals
- Historical balances:
 - Load YTD each month; reverse in the beginning of next month
 - Load YTD in the first month and PTD for each month
- Load the data in chunks; make sure the journals in the file are balanced and doesn't result in suspense or intercompany journals
- [General Ledger Data and Setup Migration Considerations](#)

Data Conversion Considerations | Guidelines

Transaction Data Migration Approach:

Load all the transaction using one dummy account.

- For example: for AP Invoices the invoice charge account and for AR invoices, the revenue account can be the dummy accounts
- Once the transactions are loaded, generate the accounting for the transaction.
- Once the journal is created in GL, reverse the journal batch to nullify the effect of migration

Use Migration control Accounts

- Create separate migration control accounts for each data objects
- Load the respective transaction using that migration control account
- Ensure that the corresponding balance in GL is migrated using the migration control account
- Ex – For AP Invoice, create a AP migration control account
- Load AP Invoice using this AP migration control account as the distribution charge account
- While loading the GL Balances, ensure that the liability balance is loaded into this AP migration control account (and not into the actual liability account)
- No need to reverse the journal created as a result of this migration

- ✓ Use parallel processing capabilities of the import processes
- ✓ Leverage DFFs to capture source object reference information like purchase order number etc.
- ✓ [How to Verify Data Conversion Results](#)

Integrations | References

- [Options for Data Migration](#)
- [ERP Cloud Implementation Leading Practices](#)
- [ERP Cloud Instance Management Plan](#)
- [ERP Cloud Guided Path and Configuration Best Practices](#)
- [Instance Strategy and Global Single Instance Guidelines](#)
- [Integrate your SaaS, On-Prem, and Cloud Applications with Oracle Integration](#)
- [Integrating and Extending SaaS using Oracle Cloud Infrastructure \(OCI\)](#)
- [Integrations: Extending and Simplifying HCM Cloud Integrations with OIC](#)
- [Best Practices: Extending CX Sales With Visual Builder](#)
- [Integration Best Practices for ERP & SCM](#)



Thank You



Appendix A | Interface Metadata (JobName, JobPackageName, Interface Options ID etc.,)

- Metadata is available in FUN_ERP_INTERFACE_OPTIONS table



- Important columns:
 - ERP_INTERFACE_OPTIONS_ID
 - UCM_ACCOUNT
 - IMPORT_JOB_NAME

Manage Enterprise Scheduler Job Definitions and Job Sets for Financial, Supply Chain Management, and Related Applications ?

Manage Job Definitions Manage List of Values Sources Manage Job Sets

Edit Job Definition

Job Definition Show More

Display Name Import Journals

Name JournalImportLauncher

Path /oracle/apps/ess/financials/generalLedger/programs/common/

Application General Ledger

Description Creates journal entries from imported information from subledgers and other source systems.

Retries 1

Job Category

Timeout Period

Job Application Name FscmEss

Job Type PlsqJobType

Procedure Name GL_INTERFACE_PKG.LAUNCH_JOURNAL_IMPORT

Default Output Format TXT

Report ID

Priority 4

Allow Multiple Pending Submissions ☒ Enable submission from Scheduled Processes

JournalImportLauncher: Parameters User Properties

Actions View Detach

Parameter Prompt	Data Type	Page Element
Data Access Set	String	Choice list
Source	String	List of values
Attribute2	String	List of values
Ledger	String	List of values
Group ID	String	Choice list
Post Account Errors to Suspense	String	Choice list
Create Summary Journals	String	Choice list

- [Oracle Integration and ERP Cloud - How to get Job Name and Parameters of ERP Bulk Import Jobs | Oracle Jack Desai's Blog](#)



Appendix B | ParameterList & ESSParameters

Tools > Scheduled Processes > Submit the job of your interest > Note down ‘All Parameter Values’

Search Results ?

View

☒ Flat List☐ Hierarchy

Actions ▾View ▾

Schedule New Process

Resubmit

Put On Hold

Cancel Process

Release Process

View Log

↺

Process ID	Name	Status
2747153	Import Journals: Child	Succeeded
2747152	Import Journals: Child	Succeeded

Import Journals, 2745826: Details

Status

Succeeded

Schedule Start

6/2/21 12:39 PM UTC

Log and Output

Attachment [ESS_L_2745826](#)

Parameters

Parameter Names with Values

Data Access Set	InFusion Ledger Set	Ledger	InFusion Ledger US	Post Account Errors to Suspense	No	Import Descriptive Flexfields	No
Source	Other	Group ID	1234567	Create Summary Journals	No		

All Parameter Values

argument1	300000206228233	argument3	300000206229310	argument5	N	argument7	N
argument2	Other	argument4	1234567	argument6	N		



Appendix C | NotificationCode & Statuses

The following table lists the execution statuses of the ESS jobs with descriptions. These statuses are returned by the getEssJobStatus operation.

ESS Job Execution Status	Description	User Action
COMPLETED	Request has completed.	This is an intermediary status and is returned when the child processes are generated. Check the status of the ESS job to ascertain whether the return status changes to SUCCEEDED or ERROR.
BLOCKED	Request is blocked by one or more incompatible requests.	Wait for the completion of the incompatible request and resubmit the process.
SUCCEEDED	Request completed and was successful.	Check the details of the completed process and proceed with any post processing.
ERROR	Request ran and resulted in error.	Download the details of the error and correct the data.
ERROR_AUTO_RETRY	Request ran, resulted in an error, and is eligible for automatic retry.	Resubmit the process after some time.
WARNING	Request ran and resulted in a warning.	Download the details of the process. Check the reason for the warnings and take the necessary action to correct the input data.
RUNNING	Request is processed.	No action.
CANCELED	Request was canceled.	Resubmit the request if required.

The following table provides information on the notification codes:

Digit Position	Digit Value	Meaning
First digit	1	E-mail notification
	2	Bell notification
	3	Email and bell notification
Second digit	0	Send in any case (import failed or succeeded)
	1	Send on import success
	2	Send on import failure



Appendix D | Callback Service

- In practice, customers will create and host a callback web service to optimally leverage the callback capabilities provided by Oracle ERP Integration Service for notification purposes.
- The customer callback web service must implement the `onJobCompletion()` operation.
- When a job completes, Oracle ERP Integration Service invokes the customer callback web service as defined in the request payload of supported operations with callback capabilities, such as the `importBulkData` operation.
- The callback response provides execution statuses and request IDs of all the applicable ESS jobs.
- The response includes the UCM document ID of the output artifacts, such as the logs of each ESS jobs, and data from the interface and error tables. This ZIP file can be retrieved using the Operation: `getDocumentForDocumentId`.

sample payload sent to callback service

```
{
  "JOBS": [
    {
      "JOBNAME": "Load Interface File for Import",
      "JOBPATH": "/oracle/apps/ess/financials/commonModules/shared/common/interfaceLoader",
      "DOCUMENTNAME": "apinvoiceimport.zip",
      "REQUESTID": "2529",
      "STATUS": "SUCCEEDED",
      "CHILD": [
        {
          "JOBNAME": "Transfer File",
          "JOBPATH": "/oracle/apps/ess/financials/commonModules/shared/common/interfaceLoader",
          "REQUESTID": "2530",
          "STATUS": "SUCCEEDED",
          "JOBNAME": "Load File to Interface",
          "JOBPATH": "/oracle/apps/ess/financials/commonModules/shared/common/interfaceLoader",
          "REQUESTID": "2531",
          "STATUS": "SUCCEEDED"
        }
      ],
      "JOBNAME": "Import Invoices",
      "JOBPATH": "/oracle/apps/ess/financials/payables/invoices/transactions",
      "REQUESTID": "2532",
      "STATUS": "SUCCEEDED",
      "CHILD": [
        {
          "JOBNAME": "Import Invoices Report",
          "JOBPATH": "/oracle/apps/ess/financials/payables/invoices/transactions",
          "REQUESTID": "2533",
          "STATUS": "SUCCEEDED"
        }
      ]
    }
  ],
  "SUMMARYSTATUS": "SUCCEEDED",
  "DOCUMENTID": "23456"
}
```

- Authentication: Supports SAML bearer token
- [ERP Integration Callback Implementation in Oracle Integration Cloud \(OIC\)](#)
- [Using External Data Integration Services for Oracle ERP Cloud \(2102800.1\)](#)



Appendix E | Job Properties File or Manifest File

- The job details include the job definition and package names, as well as the job parameters of the imported object. The following options may be used to specify the Job Details parameter associated with the **importBulkData** operation:

- Specify the Job Details parameter directly in the request payload
- Add the Job Property file as part of the data zip file
 - if data file name has “_”, job property file name should be the <datafilenameprefix>.properties;
 - if no “_” in the name, property file name should be <datafilename>.properties (DocID 2398727.1)
- Upload the Job Properties file to the UCM and add **JobDetailFileName=** in **jobOptions**

- Job property file format:

- <job package name>,<job def name>,<Manifest file name>,<job parameters (comma separated)>

Example:

- oracle/apps/ess/financials/generalLedger/programs/common,JournalImportLauncher,JournalImportTEST,1061,Payables,1,ALL,N,N,N

Multi-thread Approach

- To increase the throughput when importing data, users can specify multiple threads in the import process. It supports a maximum of 10 threads for sequential processing and a maximum of 5 threads for parallel processing. After the data file is loaded in the interface table, the import process will start batch processing based on number of job parameters records defined in a property file and job option in the payload.
- The default option is sequential and the following JobOptions property in attribute could enable parallel processing: **ExecutionMode=Parallel**
- In a job property file, you must enter multiple records with parameter values that can import data in batches either sequentially or concurrently. In sequential pattern, an import process stops when a batch fails. The remaining batch processes will not be executed and callback will include all the details including the failed process.
- The following is the sample property file of journals import where data file contains 3 different ledgers:

```
oracle/apps/ess/financials/generalLedger/programs/common,JournalImportLauncher,GL,1061,Payables,1,ALL,N,N,N
oracle/apps/ess/financials/generalLedger/programs/common,JournalImportLauncher,GL,1061,Payables,2,ALL,N,N,N
oracle/apps/ess/financials/generalLedger/programs/common,JournalImportLauncher,GL,1061,Payables,3,ALL,N,N,N
```

- ERP will import three ledgers either sequentially or concurrently depending on the “**ExecutionMode**” type. The job package and name must be same for all the records.



Appendix F | Custom ESS Jobs

[List of Values](#) for custom ESS Job parameters

- Parameters in predefined job definitions can have lists of values too, but not from list of values sources that you can manage. But if you duplicate a predefined job definition, list of values sources might be automatically created for your new job definition. You can manage those sources as you would a source that you created yourself.

✓ How to configure [dependencies](#) between parameters

- ✓ List Of Available Public Values Source For Custom Enterprise Scheduler Job (Doc ID 2071153.1)
- ✓ How To Create Custom ESS Parameter LOV Using Lookup Codes? (Doc ID 1947723.1)
- ✓ How to Create Dependent LOV Parameters for Custom ESS Jobs Using Value Sets? (Doc ID 2173057.1)
- ✓ ER - Fusion Apps - How To Create Dependent User Defined LOV In ESS Job Using SQL Query (Doc ID 2566254.1)
- ✓ Secure BU LOV at Custom BI Reports (Doc ID 2551638.1)
(FUN_ALL_BUSINESS_UNITS_V & FUN_USER_ROLE_DATA_ASGNMNTS)
- ✓ What Job Role Will Allow a Single Fusion User to View all ESS Scheduled Processes/Jobs? (Doc ID 1467664.1)

How to increment date parameter automatically for recurring runs:

- Job definition > More
- Set 'Date Increment Procedure' = NULL (internally GL_SRS_INCREMENTOR_API.INCREMENT_ESS_PARAMS proc is used)
- Set 'Date Argument Numbers' = 4,5 (parameter numbers for date parameters that need to be incremented; in this example 4th & 5th parameters are date parameters)

Appendix G | API Authentication

Inbound API calls

- Supported: **Basic-Auth, JWT & SAML** [for Cloud SSO]
 - Get public key certificate from client application (calling application) – self-signed or signed by third party
 - **In Cloud:** Security Console > API Auth > Create Oracle API Auth Provider > Choose JWT or SAML > Save > Add new certificate > import the public key certificate)
 - **In Calling App:** generate JWT (based on encoded cert fingerprint, cloud username etc.,)
 - Pass this JWT token in REST or SOAP header
 - In Cloud, this token is validated against the certificate & cloud user name
-
- [Self Service API Authentication Setup \(CCC Replay\)](#)
 - [Using OAuth when calling SaaS REST API](#)

Outbound API calls

- Supported: **JWT Claims & OAuth**
- Download public key certificate from Cloud (Security Console > API Auth > Oracle Public Certificate)
- Share the public key certificate with the target application
- Target application must import this public key certificate into its application
- In Cloud, Security Console > Create External Client Application
- Choose JWT or OAuth and provide the required parameters (if any)
- Cloud products that support out of the box integrations with third parties will generate the encoded certificate fingerprint, username etc additional parameters and make the external API call using the external client application setup)

X.509 Certificates (for HTTPS / SSL handshake)

- Created using Security Console > Certificates
- Used for: External application page integration; inbound & outbound API calls

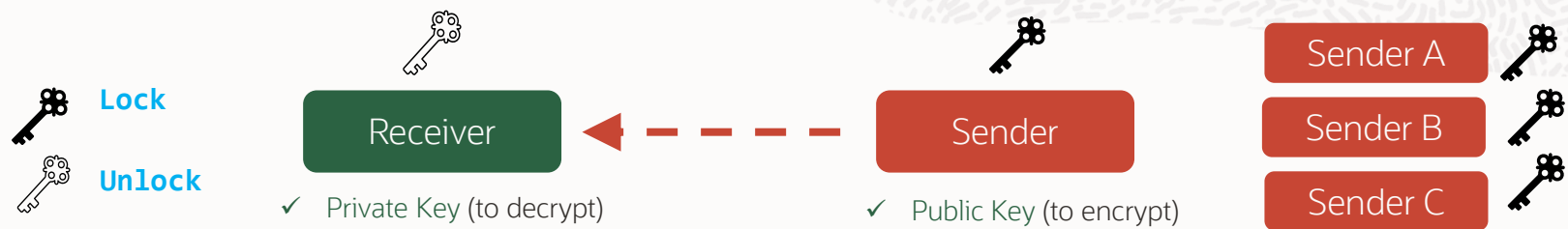
Appendix H | Whitelisting & Location Based Access Control

- [Whitelisting](#) self-service instructions
- [Location Based Access Control](#) ([Blog](#))
- LBAC Vs Whitelisting [Blog](#)
- Oracle Allow List IP's For Outbound Connectivity For Fusion Applications (Doc ID [2698639.1](#))

Appendix I | Collaboration Messaging Framework | EDI Support

- EDI is not directly supported by Collaboration Messaging Framework (CMK) [ERs 26878073, 26916900]
 - Following options can be explored:
 - Option A: Use Oracle Integration Cloud (OIC) to integrate with CMK for both inbound and outbound messaging. CMK can be configured to raise 'B2B Message Created' event which can be consumed by OIC to transform XML to EDI or vice versa.
 - Option B: Use third party service providers such as Justransform that provide EDI capabilities out of the box. These service providers deliver pre-built integrations with CMK, for all the messages mentioned above.
-
- ✓ [\[20D\] Raise a Public Business Event for Outbound B2B Messages](#)
 - ✓ [Integration Cloud: Introduction to Trading Partner Management](#)

Appendix J | File PGP Encryption & Digital Signature | Overview



Outbound

Cloud	External Party
Sign the file digitally using Cloud Private Signing Key Sign	
Encrypt the file using Third Party Public Encryption Key Lock	
	Decrypt file using Third Party Private Encryption Key Unlock
	Validate the signature using Cloud Public Signing Key Validate

Inbound

Cloud	External Party
	Sign the file digitally using Third Party Private Signing Key Sign
	Encrypt the file using Cloud Public Encryption Key Lock
Decrypt file using Cloud Private Encryption Key Unlock	
Validate the signature using Third Party Public Signing Key Validate	



Appendix J | PGP | Financials Coverage

File Based Data Import & Financial Extracts

- **Unsigned PGP** encryption only (no signature)
- [Generate](#) public & private key (**cloud-alias**)
- Import third-party public key; Security Console > Certificates > Import (**third-party-alias**)
- Encrypt & decrypt using **exportBulkData** and **importBulkData** processes respectively
- Specify aliases in **JobOptions**:

```
"JobOptions": "FileEncryption=PGPUNSIGNED, FA_ALIAS=<cloud-alias>, CUSTOMER_ALIAS=<third-party-alias>"
```

BIP Report Output

- **Unsigned & Signed PGP** encryption
- Configured using BIP > Administration page
- [How to Configure PGP in BIP for FTP](#) (DocID 1987283.1)

Payments & Cash Management

- **Signed PGP** encryption only for both inbound & outbound
- **Outbound** files: payment, positive pay & settlement batch files; **Inbound** files: bank statement & acknowledgement files
- Generated or imported using Manage Transmission Configuration page
- Stored in the wallet created by “Manage System Security Options” task
- Respective ESS jobs will encrypt / decrypt the files
- [PGP Encryption](#) and Digital Signature for Payment Files

Expenses

- **Unsigned PGP** decryption for credit card transactions file
- [Generate](#) public & private key (with EXM_PGP_KEY alias)
- Decrypted by ‘Upload Credit Card Transaction File’ process

Appendix K | Business Events

- Business Event is a definable logical occurrence in a business scenario. It can be a high-level occurrence such as payables invoice creation or a specialized event such as status change.
- If you want to perform operations in other systems based on public events in Oracle Financials Cloud, then you can use business event features. You must enable the business event if it isn't enabled by default.
- After you enable these features, the application sends a signal, along with a **payload** containing information about the public event, whenever an event occurs. The public event signal is also known as a public event.
- The application sends signals irrespective of the source or cause of the event. For example, the application sends a payables invoice creation signal when an invoice is created using a user interface or REST API.
- Integration developers and administrators can subscribe to public events from the **Oracle Integration Cloud** using the **Oracle ERP Cloud Adapter**.

- Manage Administrator Profile Values task
 - ✓ AP: Enable Business Events
 - ✓ IBY: Enable Business Events
- [Financials Business Events](#)
- [Procure to Pay – Business Events](#)
- [Common Patterns](#)